MA 324

STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS

IIT GUWAHATI

END-SEMESTER EXAMINATION

14:00–17:00 IST

MAY 8, 2026

**PART-B**

Name of the Student: Ayan Mondal

Signature of the Student: 

Roll No.: 230123012

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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| Question: | 1 | 2 | 3 | 4 | Total |
|---|---|---|---|---|---|
| Points: | 6 | 7 | 5 | 2 | 20 |
| Score: | 0 | 7 | 0 | 2 | 09 |

QUESTIONS

1. (6 points) Let

$$\boldsymbol{S} = \begin{pmatrix} S_{yy} & S_{y\boldsymbol{X}}^{T} \\ S_{y\boldsymbol{X}} & S_{\boldsymbol{X}\boldsymbol{X}} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1\times 1}$ and $\boldsymbol{X}_{q\times 1}$ based on a centered (with respect to sample mean) sample of size $n$. Prove that the squared sample first canonical correlation $r_1^2$ is identical to the coefficient of determination ($R^2$) obtained from the multiple linear regression of $Y$ on the vector $\boldsymbol{X}$.

$$\boxed{Y = X\beta + e}$$

Multiple linear Regression.

$$r_1^2 = \frac{\left[\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

We can write

$$(y_i-\hat{y}_i)+(\hat{y}_i-\bar{y})$$

$$= \frac{\left[\sum_{i=1}^{n}(x_i-\bar{x})\{(y_i-\hat{y}_i)+(\hat{y}_i-\bar{y})\}\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \frac{\left[\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\hat{y}_i)+\sum_{i=1}^{n}(x_i-\bar{x})(\hat{y}_i-\bar{y})\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \frac{\left[\cancel{\sum_{i=1}^{n}(x_i-\bar{x})e_i}^{\,0} + \sum_{i=1}^{n}(x_i-\bar{x})(\hat{y}_i-\bar{y})\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2 \sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \frac{\left[\sum_{i=1}^{n}(x_i-\bar{x})(\hat{y}_i-\bar{y})\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2 \sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \frac{\cancel{\sum_{i=1}^{n}(x_i-\bar{x})^2} \sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}{\cancel{\sum_{i=1}^{n}(x_i-\bar{x})^2} \sum_{i=1}^{n}(y_i-\bar{y})^2}$$

$$= \frac{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

or $\frac{MS_{Reg}}{SS_T} = R^2$ (Proved)

2. Suppose you are performing a factor analysis on a $4 \times 4$ sample covariance matrix $\mathbf{S}$. The eigenvalue of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.

   (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.

   (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.

   (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^{p} \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

2. (a) $\lambda_1 = 0.8$ $\lambda_2 = 2.1$, $\lambda_3 = 4.2$ & $\lambda_4 = 0.2$

$e_1$ $e_2$ $e_3$ $e_4$

As $m=2$ given so we will us largest two eigenvalues of their corresponding normalized eigen-vectors.

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$$\tilde{L} = [\sqrt{\lambda_3}\, e_3 \quad \sqrt{\lambda_2}\, e_2]$$

$$\boxed{\tilde{L} = [2.05\, e_3 \quad 1.45\, e_2]} \text{ — (a)}$$

(b) $\tilde{L}$ $l_1^T = [0.8, -0.4]$

$$\therefore \tilde{h}_1^2 = h_{11}^2 + h_{12}^2 = (0.8)^2 + (-0.4)^2$$

$$= 0.64 + 0.16$$

$$= 0.80 \text{ — (b)}$$

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(c) Sum of all estimated communities:

$$= \sum_{i=1}^{p} \tilde{h}_i^2$$

$$= \tilde{h}_1^2 + \tilde{h}_2^2 + \text{------} + \tilde{h}_p^2$$

$$= (\tilde{h}_{11}^2 + \tilde{h}_{12}^2) + (\tilde{h}_{21}^2 + \tilde{h}_{22}^2) + \text{------} + (\tilde{h}_{p1}^2 + \tilde{h}_{p2}^2)$$

$$= \sum_{i=1}^{p} (\tilde{h}_{i1}^2 + \tilde{h}_{i2}^2)$$

$$= \sum_{i=1}^{p} (\lambda_3 e_{i1}^2 + \lambda_2 e_{i2}^2)$$

$$= \lambda_3 \sum_{i=1}^{p} e_{i1}^2 + \lambda_2 \sum_{i=1}^{p} e_{i2}^2$$

As the eigen vector are normalize so if we sum all elements in a $e_i$ it will be 1

$$\therefore \sum_{i=1}^{p} e_{i1}^2 = 1 = \sum_{i=1}^{p} e_{i2}^2$$

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$$= \lambda_3(1) + \lambda_2(1) = \lambda_3 + \lambda_2$$

(Proved)

3. (5 points) Let $X_1, X_2, \ldots, X_n$ be a random sample of size $n$ from a distribution with mean $\mu \in$ and variance $\mu^2$. Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Derive a variance-stabilizing transformation $g(\overline{X}_n)$ such that the asymptotic variance of $g(\overline{X}_n)$ does not depend on $\mu$.

$$* \quad E[\overline{X}_n] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{n\mu}{n} = \mu$$

$$\mathrm{Var}(\overline{X}_n) = \mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{\sum \mathrm{Var}(X_i)}{n^2}$$

$$= \frac{n\mu^2}{n^2} = \frac{\mu^2}{n}$$

$$\therefore \frac{\overline{X}_n - \mu}{\sqrt{\mu^2/n}} \sim N(0,1)$$

$$\therefore g(\overline{X}_n) = \sqrt{\frac{n}{\mu^2}}(\overline{X}_n - \mu)$$

is a variance stabilizing transformation ~~the~~ whose asymptotic variance does not depend on $\mu$.

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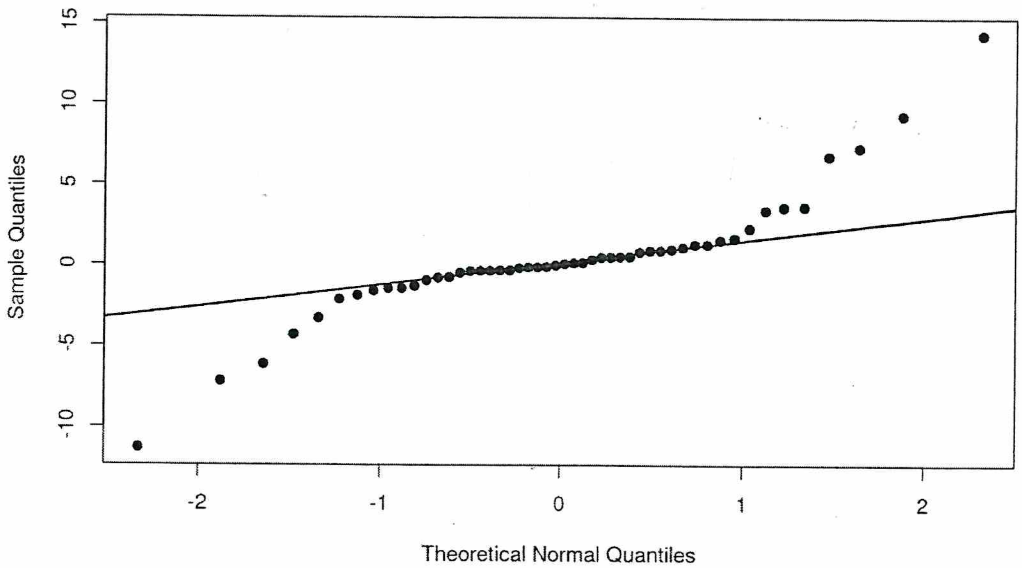
.. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.




⇒ The normal distribution doesn't follow here. The t-distribution would more accurately characterized the data here.

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Additional Pages for Answers